

School of Statistics
University of Minnesota
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Academic Appointments

IRSA Faragher Distinguished Postdoctoral Fellow, School of Statistics, University of Minnesota 2025–2027

Education

Ph.D. in Statistics (2025), University of Minnesota,
Adviser: Adam J. Rothman
M.S. in Mathematical Sciences (2020), University of Minnesota,
B.S. in Statistics (2018), Jilin University, China

Research interests

Quantum Information Science (theory, advantage, and tomography); High-Dimensional Statistics; Statistical Learning and Generative AI; Astrophysics and Cosmology; Non-Convex Optimization.

Research

— Manuscripts submitted for publication

* Undergraduate research mentee under my supervision.

19. [Hongru Zhao](#). (2026) Local Anticoncentration for Gaussian Boson Sampling via Conditional Wishart Geometry. Preprint. [Lean verified](#).
18. [Hongru Zhao](#). (2026) [Shifted Anticoncentration for Real Gram Hafnians and Symmetric Gaussian Hafnians](#). Preprint. [Lean verified](#).
17. [Hongru Zhao](#). (2026) [Uniform Hiding of Haar Block Transpose Gram Matrices](#). Preprint. [Lean verified](#).
16. [Hongru Zhao](#). (2026) [Weak Typicality of von Neumann Entanglement Entropy in Gaussian Boson Sampling](#). Preprint. [Lean verified](#).
15. [Hongru Zhao](#). (2026) [Exact Moments of Gaussian Gram Hafnians Reveal an \$n^2/\log n\$ Threshold for Weak Anticoncentration](#). Preprint. [Lean verified](#).
14. [Hongru Zhao](#). (2026) [Sharp Berry–Esseen Bounds for the Log Determinant of a Gaussian Sample Correlation Matrix](#). Preprint. [Lean verified](#).
13. [Hongru Zhao](#). (2026) [On the Log Determinant of Sample Correlation Matrices under Gaussianity](#). Preprint.
12. [Hongru Zhao](#) and Galin Jones. (2026) A Counterexample to Karlin’s Conjectured Converse for Admissibility in Exponential Families. Preprint.
11. [Hongru Zhao](#) and Huiqian Feng*. (2026) [Convex Reparameterization and Self-Concordant Algorithms for Multivariate Regression with Covariance Estimation](#). Preprint.

10. [Hongru Zhao](#), Jinwen Fu, Tuan Pham. (2025) [Convergence and Stability Analysis of Self-Consuming Generative Models with Heterogeneous Human Curation](#). Preprint.
 9. [Hongru Zhao](#), Aaron J. Molstad, and Adam J. Rothman. (2025) [Subspace Decompositions for Association Structure Learning in Multivariate Categorical Response Regression](#). Preprint.
- Peer reviewed articles
- † Authors are listed alphabetically; author order does not indicate relative contribution.
8. Xiangyu Zhang, Erik Floden, [Hongru Zhao](#), Sara Algeri, Galin Jones, Vuk Mandic, and Jesse Miller. (2026) [Validating Angular Power Spectral Models for the Stochastic Gravitational-Wave Background Without Distributional Assumptions](#). Physical Review D, 113(2), 023051.
 7. Sara Algeri, Xiangyu Zhang, Erik Floden, [Hongru Zhao](#), Galin L. Jones, Vuk Mandic, and Jesse Miller. (2026) [Testing models for angular power spectra: A distribution-free approach](#). Physical Review D, 113(2), L021306.
 6. Zhibo Zhang, Bixiong Li, Zhiwen Wang, Lianghui Li, Qingshun Nong, and [Hongru Zhao](#). (2026) [Microstructure-sensitive modeling of foam concrete based on pore-scale effect quantification and nonlinear assessment of its multifunctional performance](#). Construction and Building Materials, 537, 147017.
 5. [Hongru Zhao](#) and Xiaotong Shen. (2025) [Distributed Algorithms for High-Dimensional Statistical Inference and Structure Learning with Heterogeneous Data](#). Statistica Sinica, 38(1).
 4. Xiaou Li and [Hongru Zhao](#)†. (2025) [Globally-Optimal Greedy Active Sequential Estimation](#). IEEE Transactions on Information Theory, 71(5), 3871–3924.
 3. [Hongru Zhao](#), and Jinchao Xu. (2024) [Convergence Analysis and Trajectory Comparison of Gradient Descent for Overparameterized Deep Linear Networks](#). Transactions on Machine Learning Research, ISSN: 2835-8856.
 2. Kangsheng Liu, Zhuangyi Liu, and [Hongru Zhao](#)†. (2024) [Exponential stability of the linear KdV-BBM equation](#). Discrete and Continuous Dynamical Systems-B, 29(3), 1206-1216.
 1. Yongcheng Qi, and [Hongru Zhao](#)†. (2021) [Limiting empirical spectral distribution for products of rectangular matrices](#). Journal of Mathematical Analysis and Applications, 502(2), 125237.

Mentoring & Supervision

- Undergraduate Research
 Huiqian Feng* — Undergraduate Research Mentee (high-dimensional statistics),
 School of Statistics, UMN Fall 2025 – Spring 2026

Invited presentations

Invited Talk and Scholarly Exchange, School of Physics and Astronomy, Sun Yat-sen University, China: Statistical Inference for the Statistical Isotropy Stochastic Gravitational Wave Background (2026).

Invited Talk, Center for Applied Mathematics, Tianjin University, China: From Fisher Information to Adaptive Experimental Design: Statistical Optimality in Quantum State Tomography (2026).

Contributed Paper Presentation, Joint Statistical Meetings (JSM), Nashville, TN: Subspace Decompositions for Association Structure Learning in Multivariate Categorical Response Regression (2025).

Poster Presentation, IRSA Conference, School of Statistics, University of Minnesota: Structure Learning in Multivariate Categorical Response Regression (2024).

Colloquium Seminar, Computer, Electrical, and Mathematical Science and Engineering Division, King Abdullah University of Science and Technology, Thuwal, Saudi Arabia: Exploring the Optimization Landscape and Training Dynamics of Deep Linear Networks (2023).

Professional service

— Departmental Service

Seminar Coordinator, UMN School of Statistics Seminar Series, Fall 2025

— External Outreach and Sponsorship

Sponsor representative, WiADS Conference — Represented the IRSA; staffed the sponsor desk, and promoted IRSA consulting and statistical workshops.

— Reviewer Service

Journal of Machine Learning Research (2026); Journal of the American Statistical Association (2026); IEEE Transactions on Information Theory (2025); Electronic Journal of Statistics (2025); Annals of Applied Probability (2022); Transactions on Machine Learning Research (2024–2025).

Courses taught

— UMN-TC: Graduate Instructor

Spring 2026: STAT 4052 Statistical Machine Learning II

Fall 2025: STAT 4102 Theory of Statistics II

Spring 2025: STAT 3301 Regression and Statistical Computing

Fall 2024: STAT 4101 Theory of Statistics I

Spring 2024: STAT 3301 Regression and Statistical Computing

Fall 2023: STAT 3011 Introduction to Statistical Analysis

— UMN-TC: Graduate Teaching Assistant

Fall 2022: STAT 5101 Theory of Statistics I

Spring 2022: STAT 5102 Theory of Statistics II

Fall 2021: STAT 3021 Introduction to Probability and Statistics

Programming Skills

Lean, Mathematica, MATLAB, Python, R.